HITEC University Taxila

Department of Cyber Security

Course: Essentials of Artificial Intelligence

Lab Manual 3

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Lab Task

Task 1. Write a Python function to calculate the factorial of a number (a non- negative integer). The function accepts the number as an argument.

Input

lab3 Task 1.py > factorial

```python
def factorial(n):
    """
    Calculate the factorial of a non-negative integer n.
    Factorial of n (n!) = n * (n-1) * (n-2) * ... * 1
    """
    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers.")
    elif n == 0 or n == 1:
        return 1
    else:
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result
try:
    num = int(input("Enter a non-negative integer: "))

    fact = factorial(num)
    print(f"The factorial of {num} is: {fact}")
except ValueError as e:
    print("Error:", e)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/L
Task 1.py"
Enter a non-negative integer: 5
The factorial of 5 is: 120
PS S:\AI Lab>
```

Task 2. Write a Python function that takes a list and returns a new list with unique elements of the first list.

a) Sample List : [1,2,3,3,4,4,5,3,7,5]

b) Unique List: [1, 2, 3, 4, 5, 7]

Input

lab3 Task 2.py > unique_elements_ordered

```python
def unique_elements(lst):
    """
    Return a new list with unique elements from the given list.
    """
    return list(set(lst))

def unique_elements_ordered(lst):
    """
    Return a new list with unique elements, preserving the original order.
    """
    unique_list = []
    for item in lst:
        if item not in unique_list:
            unique_list.append(item)
    return unique_list
sample_list = [1, 2, 3, 3, 4, 4, 5, 3, 7, 5]
unique_list = unique_elements(sample_list)
print("Unique List (unordered):", unique_list)
unique_list_ordered = unique_elements_ordered(sample_list)
print("Unique List (ordered):", unique_list_ordered)
```

Output

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Task 2.py"
Unique List (unordered): [1, 2, 3, 4, 5, 7]
Unique List (ordered): [1, 2, 3, 4, 5, 7]
PS S:\AI Lab>
```

Task 3. Write a Python function to show the first ‘n’ even numbers and then return the sum and product of that even numbers outside the function.

Input

```python
def first_n_even_numbers(n):
    """
    Print the first n even numbers.
    Return the list of those even numbers.
    """
    even_numbers = []
    for i in range(1, n + 1):
        even_numbers.append(2 * i)
    print("First", n, "even numbers:", even_numbers)
    return even_numbers
n = 5
even_list = first_n_even_numbers(n)
sum_even = sum(even_list)
product_even = 1
for num in even_list:
    product_even *= num
print("Sum of even numbers:", sum_even)
print("Product of even numbers:", product_even)
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Programs/Pyt
Task 3.py"
First 5 even numbers: [2, 4, 6, 8, 10]
Sum of even numbers: 30
Product of even numbers: 3840
PS S:\AI Lab>
```

Task 4. Write a python program to match two string using functions

input

lab3 Task 4.py > match_strings

```python
def match_strings(str1, str2):
    """
    Compare two strings and return True if they match, False otherwise.
    """
    return str1 == str2
string1 = input("Enter the first string: ")
string2 = input("Enter the second string: ")
if match_strings(string1, string2):
    print("The two strings match!")
else:
    print("The two strings do not match.")
```

Output

```
Task 4.py"
Enter the first string: hello
Enter the second string: hello
The two strings match!
PS S:\AI Lab>
```

Task 5. Write a python program to name user login system using function.

Input

```
lab3 Task 5.py > login_system
def login_system(username, password):
    """
    Check if the entered username and password match the stored credentials.
    """
    users = {
        "admin": "admin123",
        "sannan": "cyber2023",
        "guest": "guest123"
    }

    if username in users and users[username] == password:
        return True
    else:
        return False
user = input("Enter your username: ")
pwd = input("Enter your password: ")
if login_system(user, pwd):
    print("Login successful! Welcome,", user)
else:
    print("Login failed! Invalid username or password.")
```

Output

```
Task 5.py"
Enter your username: sannan
Enter your password: cyber2023
Login successful! Welcome, sannan
PS S:\AI Lab>
```

Conclusion:

This lab task enhanced my understanding of Python functions and their practical use in solving problems. I implemented programs for factorial calculation, handling unique list elements,

generating even numbers with their sum and product, string matching, and a simple login system. Through these exercises, I learned how functions make code modular, reusable, and easier to manage, while also strengthening my skills in loops, conditionals, and data structures. Overall, this lab provided a solid foundation for writing structured programs and prepared me for more advanced Python concepts.